

Dialogue Ontology Relation Extraction via Constrained Chain-of-Thought Decoding

Renato Vukovic, David Arps, Carel van Niekerk,
Benjamin Matthias Ruppik, Hsien-Chin Lin, Michael Heck,
Milica Gašić

Dialog Systems and Machine Learning
Institut für Informatik
Heinrich-Heine-Universität Düsseldorf

Ontology Construction

- **Structured information**, such as *ontologies*, is crucial for scalable TOD systems and reliable LLM applications

Ontology Construction

- **Structured information**, such as *ontologies*, is crucial for scalable TOD systems and reliable LLM applications

Problems

- LLMs hallucinate on **unseen data** and do not reason
- Available reasoning data is mainly limited to logical tasks
- TOD ontologies are rare, manual construction is **expensive**

Ontology Construction

- **Structured information**, such as *ontologies*, is crucial for scalable TOD systems and reliable LLM applications

Problems

- LLMs hallucinate on **unseen data** and do not reason
- Available reasoning data is mainly limited to logical tasks
- TOD ontologies are rare, manual construction is **expensive**

We propose to **automate ontology construction** by extracting terms and relations between them!

→ Apply reasoning technique to *non-traditional* reasoning task of relation extraction (RE)

TOD Ontology Construction

"user": "am looking for a place to to stay that has cheap price range it should be in a type of hotel"
"system": "okay , do you have a specific area you want to stay in ?"

[sic!] from first MultiWOZ dialogue

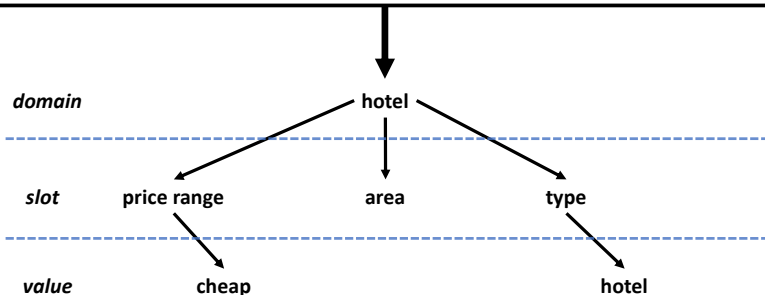
Step 1: Term Extraction

"user": "am looking for a place to to stay that has **cheap price range** it should be in a **type** of **hotel**"
"system": "okay , do you have a specific **area** you want to stay in ?"

[sic!] from first MultiWOZ dialogue

Step 2: Relation Extraction

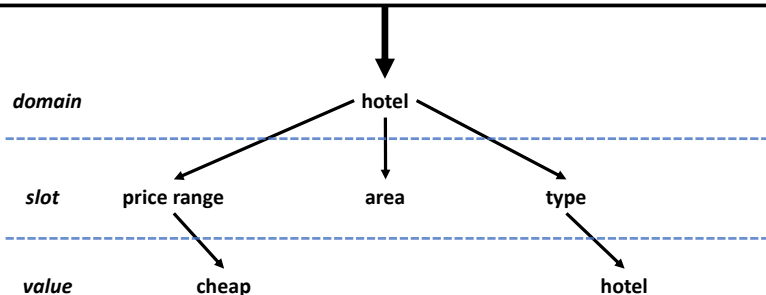
"user": "am looking for a place to to stay that has **cheap price range** it should be in a **type of hotel**"
"system": "okay , do you have a specific **area** you want to stay in ?"



[sic!] from first MultiWOZ dialogue

Step 2: Relation Extraction

"user": "am looking for a place to to stay that has **cheap price range** it should be in a **type of hotel**"
"system": "okay , do you have a specific **area** you want to stay in ?"



[sic!] from first MultiWOZ dialogue

Reasoning: "The general topic in this dialogue is the term hotel, which can hence be considered the domain. The terms price range, area and type are used as types of information relevant for hotel booking, i.e. slots. The values..."

Why not just use LLMs?

- LLMs struggle with **extracting multiple relations at once** (Lilong et al., 2024).

Why not just use LLMs?

- LLMs struggle with **extracting multiple relations at once** (Lilong et al., 2024).
- Performance is highly dependent on **pre-training data distribution** (McCoy et al., 2023).

Why not just use LLMs?

- LLMs struggle with **extracting multiple relations at once** (Lilong et al., 2024).
- Performance is highly dependent on **pre-training data distribution** (McCoy et al., 2023).
- Instruction-tuning datasets hardly cover RE (Wang, Mishra, et al., 2022)
→ LLMs **do not reason** in underrepresented tasks (Wang and Zhou, 2024)

Proposal

- Apply a technique developed for **logical reasoning** to dialogue ontology relation extraction (DORE)
 - without *prompt engineering or reasoning data*
 - Constrained chain-of-thought (CoT) Decoding

Chain-of-Thought Decoding

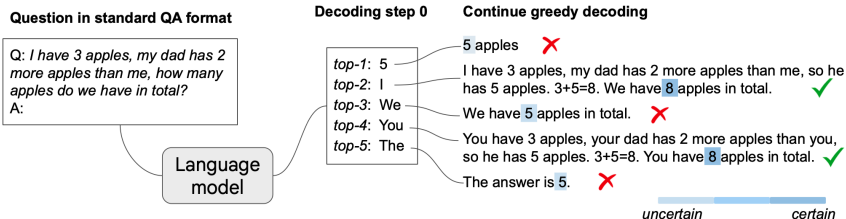


Figure: CoT-Decoding example (Wang and Zhou, 2024)

- Presence of *reasoning* is correlated with **answer confidence**
- Sample based on top- k first tokens and choose highest confidence answer
→ These completions tend to contain reasoning, since answer is not predicted greedily

CoT-Decoding for DORE

Prompt: For the listed terms in the dialogue predict the “has value” relation between slots and values in the format [slot, has value, value].

Dialogue Input:

"user": "am looking for a place to to stay that has **cheap price range** it should be in a **type** of **hotel**"

...

"user": "yes , please . **6 people 3 nights** starting on **tuesday** ."

"system": "i am sorry but i wasn ' t able to book that for you for

...

"user": "no , that will be all . good bye ."

"system": "thank you for using our services ."

Termlist:

[cheap, price range, type, hotel, 6, people, 3, nights, tuesday]

CoT-Decoding for DORE

Prompt: For the listed terms in the dialogue predict the "has value" relation between slots and values in the format [slot, has value, value].

Dialogue Input:

"user": "am looking for a place to to stay that has **cheap price range** it should be in a **type** of **hotel**"

...

"user": "yes , please . **6 people 3 nights** starting on **tuesday** ."

"system": "i am sorry but i wasn ' t able to book that for you for

...

"user": "no , that will be all . good bye ."

"system": "thank you for using our services ."

Termlist:

[cheap, price range, type, hotel, 6, people, 3, nights, tuesday]

Language
Model

Top first tokens:

- 1) [
- 2) Here
- 3) The
- 4) Sure
- 5) There
- 6) ...

CoT-Decoding for DORE

Prompt: For the listed terms in the dialogue predict the "has value" relation between slots and values in the format [slot, has value, value].

Dialogue Input:

"user": "am looking for a place to to stay that has **cheap price range** it should be in a **type** of **hotel**"
...
"user": "yes , please . **6 people 3 nights** starting on **tuesday** ."
"system": "i am sorry but i wasn ' t able to book that for you for
...
"user": "no , that will be all . good bye ."
"system": "thank you for using our services ."

Termlist:

[cheap, price range, type, hotel, 6, people, 3, nights, tuesday]

Language
Model

Top first tokens:

- 1) [
- 2) Here
- 3) The
- 4) Sure
- 5) There
- 6) ...

Greedy completions for top first tokens:

Branch 1:

[hotel, has value, price range]

...

Branch 2:

Here are the relations between slots and values: [price range, has value, cheap]

...

low confidence

high confidence



Constrained CoT-Decoding for DORE

- Only allow relation predictions that contain terms and relations given in the input

Prompt: For the listed terms in the dialogue predict the "has value" relation between domains and slots in the format [domain, **has slot**, slot].

Dialogue Input:

"user": "am looking for a place to stay that has **cheap price range** it should be in a **type of hotel**"

...

"user": "yes , please . **6 people 3 nights** starting on **tuesday** ."

"system": "i am sorry but i wasn ' t able to book that for you for

...

"user": "no , that will be all . good bye ."

"system": "thank you for using our services ."

Termlist:

[cheap, price range, type, hotel, 6, people, 3, nights, tuesday]

Greedy completions for top first tokens:

Branch 2:

The relations between slots and values in this dialogue are the following:

user
[

low confidence

high confidence



Constrained CoT-Decoding for DORE

- Only allow relation predictions that contain terms and relations given in the input

Prompt: For the listed terms in the dialogue predict the "has value" relation between domains and slots in the format [domain, **has slot**, slot].

Dialogue Input:

"user": "am looking for a place to stay that has **cheap price range** it should be in a **type of hotel**"

...

"user": "yes , please . **6 people 3 nights** starting on **tuesday** ."

"system": "i am sorry but i wasn ' t able to book that for you for

...

"user": "no , that will be all . good bye ."

"system": "thank you for using our services ."

Termlist:

[cheap, price range, type, hotel, 6, people, 3, nights, tuesday]

Greedy completions for top first tokens:

Branch 2:

The relations between slots and values in this dialogue are the following:



low confidence

high confidence



Constrained CoT-Decoding for DORE

- Only allow relation predictions that contain terms and relations given in the input

Prompt: For the listed terms in the dialogue predict the "has value" relation between domains and slots in the format [domain, **has slot**, slot].

Dialogue Input:

"user": "am looking for a place to stay that has **cheap price range** it should be in a **type of hotel**"

...

"user": "yes , please . **6 people 3 nights** starting on **tuesday** ."

"system": "i am sorry but i wasn ' t able to book that for you for

...

"user": "no , that will be all . good bye ."

"system": "thank you for using our services ."

Termlist:

[cheap, price range, type, hotel, 6, people, 3, nights, tuesday]

Greedy completions for top first tokens:

Branch 2:

The relations between slots and values in this dialogue are the following:



low confidence

high confidence

Experimental Setup

- Use *labelled seed* dataset to predict relations on *unseen target* data
- Seed data setups:
 - One-shot in-context learning
 - Fine-tuning
- Decoding approaches:
 - Greedy decoding
 - **Constrained CoT-decoding**

Data-sets

- Seed dataset: [MultiWOZ](#) (Budzianowski et al., 2018)
 - 10,000 dialogues covering **7 domains**
↪ restaurant, hotel, attraction, taxi, train, hospital, police
 - We use the training set for training one-shot example selection.
- Target dataset: [Schema-Guided Dialogue](#) (SGD) (Rastogi et al., 2020)
 - SGD test split is used for evaluation, which contains 4,201 dialogues and **18 domains**.
↪ travel, events, payment, media, restaurants, weather, ...

SGD Test Set Results with Gemma 2B Instruct

Approach	F1-Score	Precision	Recall
<i>One-shot example from MultiWOZ</i>			
<i>Baseline</i>	7.4	8.8	6.4
<i>Fine-tuning on MultiWOZ</i>			
<i>Baseline</i>	10.9	6.8	28.8

SGD Test Set Results with Gemma 2B Instruct

Approach	F1-Score	Precision	Recall
<i>One-shot example from MultiWOZ</i>			
<i>Baseline</i>	7.4	8.8	6.4
+ constrained CoT decoding	9.2	6.4	15.9
<i>Fine-tuning on MultiWOZ</i>			
<i>Baseline</i>	10.9	6.8	28.8
+ constrained CoT decoding	13.7	9.8	23.0

⇒ **Target data performance** is improved by constrained CoT-decoding

SGD Test Set Results with Gemma 2B Instruct

Approach	F1-Score	Precision	Recall
<i>One-shot example from MultiWOZ</i>			
<i>Baseline</i>	7.4	8.8	6.4
+ constrained CoT decoding	9.2	6.4	15.9
<i>Fine-tuning on MultiWOZ</i>			
<i>Baseline</i>	10.9	6.8	28.8
+ constrained CoT decoding	13.7[‡]	9.8	23.0 [‡]
<i>Upper Bounds using SGD Training Data</i>			
One-shot example from SGD + separate relation prediction	12.9	10.7	16.4
Fine-tuning on SGD	37.3	27.9	57.2

- ⇒ **Target data performance** is improved by constrained CoT-decoding
- ⇒ Fine-tuned model **outperforms target one-shot** approach (highlighted by [‡])

One-shot Example

PROMPT: <EXAMPLE> Predict the relations between equivalent terms in the form [term1, refers to same concept as, term2] as in the example given above.

RESPONSE:

BRANCH 0:

****Domain:**** Hotel
****Slot:**** Type of hotel
****Value:**** Price range
- [price per night, refers to same concept as, 135 carlingview drive etobicoke]

....

BRANCH 1:

Synonyms/Semantically Equivalent Terms:
****Domain:**** Hotel
- best western plus toronto airport carlingview hotel
-

BRANCH 2:

1. ****price range**** [1, refers to same concept as, price]...

BRANCH 3:

• ****price range**** [1, refers to same concept as, price]
• ...

BRANCH 4 (highest confidence):

Here are the predicted synonyms or semantically equivalent terms from the provided dialogue:

****Price range:****

- [price per night, refers to same concept as, cost]

....

****Day:****

- [tuesday next week, refers to same concept as, check in date]

- Branch 4 contains an explanation
- It is chosen as the final response based on confidence

One-shot Negative Example

PROMPT: <EXAMPLE> Predict the relations between equivalent terms in the form [term1, refers to same concept as, term2] as in the example given above.

RESPONSE:

BRANCH 0:

Domain: Hotel
Slot: Price range
Term: Price...

BRANCH 1:

Synonyms/Semantically Equivalent Terms:
...

BRANCH 2 → with reasoning:

The following terms are synonyms or semantically equivalent:

- [6 : 30 pm, refers to same concept as, 18 : 30]
- [4 : 30 pm, refers to same concept as, 16 : 30]

BRANCH 3 → highest confidence :

1. **price range** [fiddler, refers to same concept as, price]
2. **reference number** [fiddler, refers to same concept as, reference]

...

BRANCH 4:

* **price range** [fiddler, refers to same concept as, price]

- Although branch 2 contains an explanation branch 3 is chosen ultimately
- In the fine-tuned model no explanations are observed
- For other relations there are shorter explanations, e.g. “The predicted relations are: ...”

Conclusion

- Ontologies are important parts of TOD systems, but manual construction is expensive
- Ontology RE is an **underrepresented** task challenging LLM performance

Conclusion

- Ontologies are important parts of TOD systems, but manual construction is expensive
- Ontology RE is an **underrepresented** task challenging LLM performance

Constrained CoT-Decoding

- Enables a source fine-tuned model to outperform a target one-shot model
- Adds textual explanations for non-fine-tuned models

Conclusion

- Ontologies are important parts of TOD systems, but manual construction is expensive
- Ontology RE is an **underrepresented** task challenging LLM performance

Constrained CoT-Decoding

- Enables a source fine-tuned model to outperform a target one-shot model
- Adds textual explanations for non-fine-tuned models

Future Work

- Induce global relations between terms from different dialogues
- Reduce the inference cost of CoT-decoding
- Better criteria for sampling and choosing the final prediction

Acknowledgements



European Research Council

Established by the European Commission



Alexander von
HUMBOLDT
STIFTUNG

Ministerium für
Kultur und Wissenschaft
des Landes Nordrhein-Westfalen



hhu.



Thank you for your interest!

References I

-  Budzianowski, Paweł, Tsung-Hsien Wen, Bo-Hsiang Tseng, Iñigo Casanueva, Stefan Ultes, Osman Ramadan, and Milica Gašić (Nov. 2018). “MultiWOZ - A Large-Scale Multi-Domain Wizard-of-Oz Dataset for Task-Oriented Dialogue Modelling”. In: *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Brussels, Belgium: Association for Computational Linguistics, pp. 5016–5026. DOI: 10.18653/v1/D18-1547. URL: <https://aclanthology.org/D18-1547>.
-  Lilong, Xue, Zhang Dan, Dong Yuxiao, and Tang Jie (2024). “AutoRE: Document-Level Relation Extraction with Large Language Models”. In: *Preprint, arXiv:2403.14888*.
-  McCoy, R. Thomas, Shunyu Yao, Dan Friedman, Matthew Hardy, and Thomas L. Griffiths (2023). “Embers of Autoregression: Understanding Large Language Models Through the Problem They are Trained to Solve”. In: *Preprint arXiv/2309.13638*. URL: <https://api.semanticscholar.org/CorpusID:262464572>.

References II

-  Rastogi, Abhinav, Xiaoxue Zang, Srinivas Sunkara, Raghav Gupta, and Pranav Khaitan (2020). “Towards Scalable Multi-domain Conversational Agents: The Schema-Guided Dialogue Dataset”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 34.05, pp. 8689–8696. DOI: [10.1609/aaai.v34i05.6394](https://doi.org/10.1609/aaai.v34i05.6394).
-  Wang, Xuezhi and Denny Zhou (2024). “Chain-of-Thought Reasoning Without Prompting”. In: *Preprint, arXiv:2402.10200*.

References III



Wang, Yizhong, Swaroop Mishra, Pegah Alipoormolabashi, Yeganeh Kordi, Amirreza Mirzaei, Atharva Naik, Arjun Ashok, Arut Selvan Dhanasekaran, Anjana Arunkumar, David Stap, Eshaan Pathak, Giannis Karamanolakis, Haizhi Lai, Ishan Purohit, Ishani Mondal, Jacob Anderson, Kirby Kuznia, Krma Doshi, Kuntal Kumar Pal, Maitreya Patel, Mehrad Moradshahi, Mihir Parmar, Mirali Purohit, Neeraj Varshney, Phani Rohitha Kaza, Pulkrit Verma, Ravsehaj Singh Puri, Rushang Karia, Savan Doshi, Shailaja Keyur Sampat, Siddhartha Mishra, Sujan Reddy A, Sumanta Patro, Tanay Dixit, and Xudong Shen (Dec. 2022). “Super-NaturalInstructions: Generalization via Declarative Instructions on 1600+ NLP Tasks”. In: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Ed. by Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang. Abu Dhabi, United Arab Emirates: Association for Computational Linguistics, pp. 5085–5109. DOI: 10.18653/v1/2022.emnlp-main.340. URL: <https://aclanthology.org/2022.emnlp-main.340>.